GraphIV experiment audit — plain-language guide

what each experiment does, what it found, and whether a rerun buys anything

2026-07-15, updated 2026-07-17

One entry per experiment. Rerun cost classes: **SQL** (minutes, no GPU), **battery** (27-topic single-pass, ∼25 min), **deep** (8-topic deep panel, GPU hours). Entries marked [**stale**] measured a configuration that no longer exists: since the suite last ran we shipped (a) the `ef_search` fix + ancestor floor 0.35→0.30 + landmark budget m 40→60 package (02-07), (b) reader fusion as default (02-05), (c) asym as the only prod embedding space, and (d) the 07-15 corpus delta (+27,131 papers, +1.74M citations, 595k newly parsed).

07-17 additions: embedding is now served by the `llama-embed` endpoint (parity cosine > 0.999, KNN identical — no experiment invalidated); the trajectory sampler is prod selection in the MCP serve path; and Section 4 is new: the survey-mined silver-qrels benchmark (external evaluation instrument) with its first four studies.

1 Preprocessing (01-*) — all SQL-cheap, descriptive

01-01 resolver quality

Each bibliography entry is resolved to an arXiv id through a ladder of matchers: literal link → regex → DOI → journal-ref → exact title → fuzzy title. This counts which rung resolved each of the 177M citation occurrences, by field and era. Found: 40.2% of citations resolve to arXiv at all (the rest cite journals/books outside the corpus); ∼80% of resolutions come from the exact rungs; the fuzzy tail is bounded at ∼11%. The graph is not built on guesses.

Rerun now? Corpus grew by 595k parsed papers today; numbers drift. Cheap; refresh for any public table.

01-02 labeling quality

Every citation edge is tagged with *why* the citing paper cites (extends/uses/compares/background/critiques/motivates) plus polarity. A 26B teacher labeled 100k samples; a DistilBERT student distilled from it labels the full 65M+ resolved edges. This measures the student on held-out teacher labels: polarity essentially solved (acc 0.95); intent acc 0.84, macro-F1 0.72; the rare `critiques` class lags (0.58).

Rerun now? Frozen artifacts; a rerun reproduces itself. Only meaningful after retraining the student.

01-03 graph structure

A census of the resolved intent-typed graph: 36.7M distinct edges over 3.07M papers, degree distributions, growth, top-cited list. Sanity anchor: in-degree ranking recovers the canonical landmarks (ResNet/BERT/Transformer). Documents the heavy-tailed hub problem that motivates "induce within a topical pool."

Rerun now? +1.7M edges today; cheap census refresh.

01-04 clustering comparison

The corpus topic tree is built by recursive bisection of the embedding cloud with a coherence early-stop. Pitted against k-means / DBSCAN / HDBSCAN on a common 30k sample: HDBSCAN dumps 59% of papers into a noise bucket, DBSCAN collapses to one blob; the tree places every paper (0% noise) at density-adaptive grain. Silhouette is the wrong lens at this grain.

Rerun now? Algorithm-choice conclusion, not data-dependent. No.

01-05 topic tree

Shape census of the shipped tree: 3,582 leaves over 2.92M papers, leaf-size/depth distributions, coherence-vs-size scatter. Confirms the early-stop keeps leaves tight.

Rerun now? The tree is frozen *by design* and now ends ∼180k papers before the corpus does; the interesting action is a phase-5 `--include-force` rebuild decision, not this census.

2 Retrieval (02-*) — picking the papers a review is built from

02-01 subgraph retrieval

The core recipe: embed the question; pull a candidate pool (vector KNN fused with lexical BM25-style match); add each pool paper's most-cited ancestors; keep only citation edges inside the pool (*induce*); rank by personalized PageRank

$$\pi \;=\; (1-\alpha)\, e_q \;+\; \alpha\, P^{\top}\pi, \tag{1}$$

where e_q concentrates restart mass on query-similar nodes. Tested against PCST and similarity-BFS selectors on expert-canonical landmark sets: recall 0.742 of canonical landmarks, 0.917 of the foundational spine, against a 0.85 pool ceiling — dominates every recall axis; PCST wins only connectivity (a presentation concern).

Rerun now? [**stale**] Ran under the `ef_search`=40 cap and pre-fusion, i.e. handicapped; a rerun on the fixed stack should score *higher*. Rerun if this number goes in the paper.

02-02 relevance judge

Recall cannot see junk. An LLM labels every selected landmark's role (foundational/core/frontier/peripheral/unrelated) across 27 topics: ∼78% on-topic, and the diagnosis that the ∼22% junk is famous cross-field hubs whose in-degree (5.4, as high as core papers) makes them invisible to PPR as noise.

Rerun now? [**stale**] Historical "before" picture; 02-05 measured the same quantity at 0.945→1.0 after the floor + fusion shipped. Keep as history.

02-03 ancestor gate

The junk enters through one door: ancestor injection. Sweep a cosine admission floor

$$\text{admit ancestor } a \iff \cos\big(u_q, u_a\big) \geq \tau, \qquad \tau \in \{0.25, \ldots, 0.45\}, \tag{2}$$

on both precision (27 topics) and gold recall (3 topics). Recall is flat on $[0.25, 0.35]$ (0.808, spine 1.0) while precision rises monotonically ⇒ shipped $\tau = 0.35$ as the max-precision edge of the plateau; $\tau \geq 0.40$ cuts genuinely foundational low-cosine papers (recall collapses to 0.522 at 0.45).

Rerun now? [**stale**] [**stale**] Superseded twice: 02-07 moved production to $\tau = 0.30$ (the reader now does the junk-filtering job) and the sweep ran under the ef=40 bug. Methodology stands; number retired.

02-04 selection flags

Given a pool, pick the final set: fixed top-50 vs cosine-gated vs LLM-filtered (same count) vs LLM-filtered-and-bigger, scored by per-paper LLM grades in $[-2, 2]$. Filtering lifts both sum (+42%) and mean (+40%); going bigger lifts only sum while the mean saturates at ∼0.88 — the same place the compose judge saturates. Filter ≫ count.

Rerun now? Absorbed into the fusion design; historical.

02-05 reader fusion

Measure-gate for the shipped default. The production reader grades every candidate (6-class role + relevance $r \in [-2, 2]$, cached by question hash); ranking fuses semantics and grade:

$$\text{score}_i \;=\; \cos\big(u_q, u_i\big) \cdot \frac{r_i + 2}{4}, \qquad r_i = -2 \Rightarrow \text{excluded from landmarks, kept as graph connector.} \tag{3}$$

Gate: gold recall (grader-independent) must not regress. Result: recall identical (0.808 / spine 1.0), unrelated landmarks 5.7% → 0%, freed slots shift to core (0.270 → 0.307) and frontier (0.224 → 0.257). Pareto win; shipped as `fuse: true`.

Rerun now? Current, load-bearing default. Rerun only to refresh numbers on the grown corpus.

02-06 selection objective

PPR itself on trial (motivated by real misses: SimPO/KTO never became landmarks). Six objectives over an identical front-end: PPR vs facility location $\max_S \sum_{p \in \text{pool}} \max_{s \in S} \text{sim}(p, s)$, MMR, cluster-proportional, fused-topk, hybrid. Verdict: **PPR vindicated** — gold recall 0.753/0.917 vs 0.30–0.44 for every set-coverage arm (facility 0.436; re-checked on uncapped pools: 0.297). The rich-get-richer bias *is* the mechanism that recovers foundational lineage; precision is already handled by fusion (relevance 1.0 on every arm). Autopsy of real misses ⇒ three causes: pool-miss, lukewarm grades on question-named entities, budget pressure — none fixed by the objective.

Rerun now? Current; verdict strengthened on the fixed stack. Do not relitigate *vs set-coverage*. Note 07-17: this verdict is about objectives; the magnet-walk trajectory hybrid (PPR scaffold + 8 consensus-walk slots, a different successor) now beats pure PPR 6/6 on the external survey benchmark (04-04) and is prod selection in the MCP path.

02-07 retrieval breadth (the bug + the coupled package)

pgvector's HNSW returns at most `hnsw.ef_search` rows regardless of `LIMIT`:

$$|\text{KNN}(q, k)| \;\le\; \min(k,\ \texttt{ef_search}), \qquad \texttt{ef_search}_{\text{default}} = 40, \tag{4}$$

and the parameter was never set — every vector KNN since the PG consolidation returned ≤ 40 rows; `n_pool`=400/600 was fiction and lexical RRF silently carried retrieval (fatal for long questions: litmus pools 60–137). Fix: `SET hnsw.ef_search = 1000` in `Store.__init__`. Then the coupled package: ef alone *drops* landmark hits (bigger pools make m=40 more competitive), so ship {ef 1000, floor 0.30, m=60} together; floor 0.30 at m=40 dilutes gold (0.753 → 0.711) — never the floor without the budget. Grading cost scales with m only (candidates = $4m$), not pool size. Result: litmus pool-hit 0.857 → 1.0, landmark-hit 0.524 → ∼0.62, gold gate held 0.753/0.917; m=80 documented as the paid tier. BYOL is the emblem: vocabulary-shift foundational paper, KNN rank > 1000, reachable only through the ancestor channel, blocked by $\tau = 0.35$ by 0.010.

Rerun now? THE regression harness. Rerun after any retrieval change and to refresh on the +27k corpus.

3 Generation (03-*) + flagship

03-01 judge validity

Validates the measurement instrument: plant known defects, check detection per axis; test position and verbosity bias. Sound: 100% on known facts, catches coverage/grounding degradation, no biases; lineage is the weakest axis. Rule: trust grounding/coverage and large effects; asterisk tiny lineage margins.

Rerun now? Judge gained retrieval-context + dated preambles in July; a light re-validation is defensible, not urgent.

03-02 component ablation

Knock out one component at a time (PPR→in-degree / no-cluster / no-graph / no-fulltext) on the 27-topic battery, reading the full metric vector + 4-axis judge + faithfulness tripwire. The suite's founding lesson: *read the vector, never one number* — `no_graph` posts the best frontier count while zeroing edges, recall, and judge score.

Rerun now? [**stale**] Ran on the pre-fusion, pre-ef-fix, sym-era stack. The paper-grade ablation table wants a rerun on the current stack.

03-03 embedding format

Query-encoding shootout, end-to-end: {sym, asym} × {HyDE, raw}. `asym_raw` ties for best recall (0.81 / spine 1.0) at no extra cost; HyDE only rescues the symmetric space. Shipped — and it is why asym is now the prod space.

Rerun now? Settled; re-open only post-LoRA (the v3 training gates on this comparison anyway).

03-04 recency knob

Sweep the lineage↔frontier dial 0.0–0.8 (frontier slots reserved newest-cohort-first; age-normalized score $\approx$ citations per year, $c/(y_{\text{now}} - y + 1)$). Frontier landmarks climb monotonically 3.7 → 30.8 at constant grounding: a true control surface with no universal optimum — correctly a per-query knob.

Rerun now? Structural behavior; no.

03-05 frontier pull

The topic tree predates the corpus delta, so tree-anchored retrieval can never surface post-tree papers. Toggle the tree-independent, year-filtered vector pull: without it, *zero* 2026 papers ever appear; with it, ∼1.9/topic at no cost. A correctness fix, shipped on.

Rerun now? More relevant after today: the un-treed delta grew by 27k and this channel is what surfaces it. Cheap confirm-run recommended.

03-06 landmark selection

Older generation-side sibling of 02-04 (40-paper scale): the win is the relevance filter, not the count; `llm40` wins at zero size increase; bigger adds nothing.

Rerun now? Redundant with 02-04/02-05; historical.

03-07 graph representation

How the subgraph is serialized for the writer: structure axis (flat/adjacency/DOT/subtree/ JSON/matrix), then role tags, extracted content, aliasing layered on the winner. Production digest = adjacency list + extracted content + roles + aliases; extracted content is the single biggest lever (0.75 pairwise net); winner tops every axis (4.906, pairwise 7/1/0) at the fewest tokens.

Rerun now? Current and load-bearing for the writer; rerun only if the digest changes.

03-08 judge loop

Test-time compute: cap judge-driven gap-expansion rounds at $\{1, 2, 4, 6\}$. Rounds double landmarks (80 → 170) and frontier coverage (37 → 74); gold recall 0.81 → 0.96; pairwise rounds-6 vs rounds-1 = 6/0/2 with zero axis losses. The thinking judge self-stops (∼3.4 rounds at cap 6), so the cap is a ceiling — shipped as the effort tiers.

Rerun now? Current deep-path behavior; deep-cost; rerun for the paper only.

03-09 critique depth

The documented null: per-paper full-PDF critiques at $k \in \{0, 4, 8\}$. k=4 nudges recall (0.88 → 0.92) and grounding; k=8 gives it all back and faithfulness slips monotonically (0.99 → 0.95 → 0.94); pairwise $k8$ vs $k0$ = 2/2/4. The deep loop's win is the rounds (03-08), not depth. Kept k=4.

Rerun now? A null that saved a cost sink; leave it.

03-10 vision figures

Toggle the figure gallery (vision model reads landmark PDFs, extracts diagrams/plots, captions verified against the image). Text judge unchanged (it cannot see images); the gallery adds ∼13 verified figures at 0.91 caption-faithfulness. Additive enrichment, shipped as an option (prod now also role-gates figures — slightly stricter than the measured config).

Rerun now? Current; only if you want the role-gated config re-measured.

03-11 hub suppression

Keep Attention/BERT/Adam from colonizing niche reviews (Attention: 27/27). Swept four data-derived genericness signals (lift, intent background-ratio, `pop_bg`, `pr_ratio`): no structural silver bullet, because a hub genuinely *used* in a niche cannot be removed without recall cost — "generic here" is partly narrative. Shipped fix lives in the writer (compose guard), A/B-validated: generic-hub mentions 3/1/3/1 → 0/0/0/0 on the test topic with niche coverage preserved.

Rerun now? Compose guard is current prod; the sweep half is research context.

deep-research (flagship)

The holistic before/after: old single-pass default vs the full deep pipeline, 8 topics, judged pairwise on *full untruncated* reports by three disjoint judge families: 8/0/0 on overall and grounding, unanimous (96 axis-verdicts, zero baseline wins; sign test $p \approx 2 \cdot (1/2)^8 \approx 0.008$ per judge). Objectively: 9.0 2026-papers/topic vs 0, 2.4× frontier coverage, 1.5× grounded citations, 13 verified figures. An earlier 16k-char judge cap saw half of each deep report and reported a softer 5/2/1 — the truncation artifact worth remembering.

Rerun now? "After" side already re-ran on the current stack (h2h); "before" is frozen by definition. Refresh only for headline numbers.

4 External evaluation: the survey-qrels benchmark (04-*, NEW 07-17)

Motivation (user call: "hand-curated gold is unscientific"): the 27-topic battery holds only 19 canonical + 13 spine ids *total* — a regression gate, not a precision/recall claim. The field's answer (probed with GraphIV itself: LitSearch, PaSa, "citation lists are not a ground truth") is reference-list silver gold + LLM adjudication. Scripts live in the session scratchpad (`survey_qrels.py`, `score_survey.py`, `recall_at_freq.py`, `compare_ablate.py`, `traj_vs_ppr.py`); the retrieval cap flag is in the repo (`--set-max-date` / `--set-exclude`).

04-01 survey-qrels pilot

A survey paper is a human expert who already did our task: read a topic, picked the papers that matter. So: take a survey's resolved reference list from our own citation table as the answer key (80–250 in-corpus ids each), derive a question from the survey, hide the survey, ask GraphIV, count how many refs come back in the pool. Weight by how often the survey *mentions* each ref (the mention rows already exist in the citation table): one-off cites are background noise; refs mentioned ≥ 3 times are what the survey is about. Report the whole recall-vs-mention-frequency *curve*, not one number. Pilot (3× Jan-2024 CS surveys, fast tier): all curves rise monotonically and hit 1.0 at $N \geq$ 5–8 mentions; misses concentrate in the 1-mention tail. Design lessons: derive questions from the survey *abstract* (title-only under-scopes — the 3DGS misses were

avatars/SLAM subtopics the narrow question never asked about); survey selection uses citation count as a *floor* (∼15–20) + subfield×year stratification, never top-cited ranking (age / mega-topic / hub-reward bias).

Rerun now? Pilot done; the ∼20-survey stratified battery is the next build.

04-02 time-capped retrieval ("no newer than the survey")

User-mandated: corpus *and clock* are frozen at the survey's publication month (`--set-max-date YYMM`: a Store-level filter across all six pool channels + deep-loop gap injection; `now` is anchored so per-topic frontier detection time-travels), and the survey itself is blocklisted (`--set-exclude` — otherwise citation drill walks the survey's own reference edges, i.e. copies the answer key; it leaked into the pool in every uncapped pilot run). Validation (diffusion-SR): leak gone, 0 postdating landmarks, core recall 0.842 capped vs 0.895 uncapped — the delta is two *pre-cap* papers reachable only via post-cap intermediaries, exactly the contamination the cap removes. None of the comparable works (LitSearch / PaSa / 2605.29234) caps explicitly: a claimable methodological novelty.

Rerun now? Shipped in the repo, both run paths, lint-clean.

04-03 does the benchmark discriminate?

A benchmark is only an instrument if it separates configs we understand. Ablations through the capped diffusion-SR survey: `no_graph` leaves pool metrics byte-identical (correct null — it cannot touch the pool) while landmark-core drops 0.263 → 0.158 (graph ranking ≈40% of core selection); `no_frontier_pull` drops tail recall 0.518 → 0.473 (detected); `no_cluster` *raised* R@2/R@3 (0.65 → 0.775 / 0.842 → 0.895) on a smaller pool — the topic-tree refinement may be evicting mid-importance refs. One-topic surprise, not a conclusion; the battery decides. Pool metrics are deterministic (no LLM in pool assembly), so pool deltas are real.

Rerun now? Instrument validated on one topic; the `no_cluster` flag is an open battery question.

04-04 trajectory vs PPR on the external instrument

First real client of the benchmark: does the magnet-walk sampler beat PPR on qrels neither was tuned against? The first attempt was **invalid** and taught the tier rule: fast tier skips reader grading ⇒ the sampler's rel≥ 2 magnet set is empty ⇒ it silently degrades to a blind walk (it swapped Score-SDE/ADM/DDIM out for 2016 CNN papers — the degraded selector, not the algorithm). Selection A/Bs must run standard/deep; fast is pool-metrics-only. Suggested guard (not yet coded): warn + fall back to `ppr` when `rank=trajectory` sees no grades. Corrected run (standard, m=80, grades cached so both arms see identical pools + grades): **trajectory wins landmark core-ref recall 6/6 cells** across the three surveys — lm@3: 0.526/0.421, 0.304/0.261, 0.381/0.333; lm@1 a wash; rep@3 tied except a 3DGS dip (0.130 vs 0.261, compose-stage, watch in the battery). Independent-instrument validation on top of the internal-battery Pareto.

Rerun now? Done; the +1/ + 2-paper margins need the 20-survey battery for intervals.

04-05 (planned) full battery + adjudication + human calibration

(a) ∼20 surveys, citation-floor + stratified, abstract-derived questions via the map LLM, all capped, both ranks, standard tier (∼2 overnights). (b) LLM adjudication of uncited landmarks by a judge *disjoint* from the pipeline's reader (GLM / GPT-OSS, not Gemma — no grading your own homework); per 2605.29234 only ∼51% of human citations clear a neutral relevance bar, so "not in refs" ≠ wrong. (c) Doomscroll labeler: ∼100–200 (question, paper) pairs labeled by the user once; if judge–user agreement matches human–human agreement (the Bing protocol), the judge scales to the rest and the paper gets an inter-annotator number.

5 Rerun shortlist

priority	experiment	why
high	04-05 battery	turns 04-01/04-04's +1/+2-paper margins into intervals; the paper's eval section
high	02-07	it *is* the regression harness; refresh on the +27k corpus
high	03-05	directly validates that the delta is retrievable end-to-end
medium	03-02	the paper-grade ablation, on a dead stack; consider re-running *through* 04 (capped surveys) instead of the internal battery
medium	02-01	headline recall number, measured under the `ef_search` handicap
medium	01-01 / 01-03	cheap SQL censuses; numbers drifted with the 07-15 ingest
low	03-01	light re-validation after the judge's July prompt changes
skip	02-02 / 02-03 / 02-04 / 03-06	superseded; keep as history
skip	01-02 / 01-04 / 01-05 / 03-04 / 03-09	frozen artifacts / structural nulls